



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Odd Semester)

Active Learning Activity- Unit-I- Introduction to Microwave Systems and Antennas

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC8701 Antennas and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Link budget and link margin	One Minute Paper	03.08.2023 & 5 Minutes

Link budget and link margin - One Minute Paper



Signature of the Faculty member

HOD/ECE



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Active Learning Activity- Unit-II- Radiation Mechanisms and Design Aspects

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC8701 Antennas and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Microstrip antennas	Mindmap	04.09.2023 & 10 Minutes

Microstrip antennas - Mindmap



Signature of the Faculty member

HOD/ECE



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Active Learning Activity- Unit-III- Antenna Arrays and Applications

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC8701 Antennas and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Uniformly spaced arrays with non-uniform excitation amplitudes	Open Book Test	27.09.2023 & 25 Minutes

Uniformly spaced arrays with non-uniform excitation amplitudes- Open Book Test



Signature of the Faculty member

HOD/ECE



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Academic Year 2023 – 2024 (Odd Semester)

Active Learning Activity- Unit-V- Microwave Design Principles

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC8701 Antennas and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Microwave Oscillator Design	Think Pair Share	01.11.2023 & 10 Minutes

Microwave Oscillator Design - Think Pair Share



Signature of the Faculty member

HOD/ECE



Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC8701 Antennas and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Innovative Practice Description

- **Unit / Topic: Unit IV / Power Divider and Magic Tee**
- **Course Outcome: CO4** The students will be able to explain the basic concept of various microwave devices.
- **Topic Learning Outcome: TLO11** The students will be able to describe the basic principles of Microwave Passive components
- **Activity Chosen: Flipped Classroom**
- **Justification:**

Power divider and the magic tee are the passive microwave devices which can be understood easily. Flipped classroom practice was chosen because the topic is easy and interesting for the students to refer by themselves.

- **Time Allotted for the Activity:** 50 minutes

- **Details of the Implementation:**

The students were given with the reference materials to get prepared. The students were asked to make a group among them. The questions were asked from the topic they have prepared. As it is power divider and Magic Tee, most of the questions were related to the ports. As a group, they have discussed and came up with answer for the questions. After that, they were asked to get the class back together to share the individual's group work with everyone. While discussing like this, they can get an in-depth knowledge about the topic and they can clearly express their ideologies related to that topic.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO8	PO9	PO10	PSO3
CO4	3	2	3	3	3	2

- **PO / PSO mapped:**

Innovative practice	PO1	PO8	PO9	PO10	PSO3
	3	3	3	3	2
Justification for correlation	Electromagnetic Fields fundamentals are applied	Group activity which requires inter-personal ethics	Needs individual and team work	Effective communication is required	Students involves design and development of microwave passive devices

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

1. They can get engaged in class for the entire 50 minutes.
2. The activity acts as a better platform for discussion.
3. It is quite interesting.

❖ ***Benefit of the practice:***

1. They will get exposed to text books and reference materials.
2. Better interaction and communication due to group activity.
3. All the students can effectively take part in the activity and the idea about the topic has been reached more students.

❖ ***Challenges faced in implementation:***

1. To organize the activity little more time has been consumed than expected.

References:

1. <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model>

**Signature of Faculty Member****HOD**